

Report On
Vulnerability Assessment and Attack Strategy
Department of Information Technology
By

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Q1

An e-commerce website was attacked during a festival sale. Later investigation revealed weak password policies and outdated software. Analyze the planning mistakes that allowed the attack to succeed. Explain how vulnerability identification and timing strategy played a role.

During a festive sale, the e-commerce website became an easy target because of poor security planning and weak vulnerability management.

(i) Planning mistakes that allowed the attack

- Weak password policy
 - Users and administrators using simple or repeated passwords.
 - Lack of multi-factor authentication increased risk of unauthorized access.
- Outdated software
 - Old version contained known security flaws.
 - Security patches were not installed on time.
- Lack of regular security audits
 - Vulnerabilities were not identified before the attack.
- Poor monitoring system
 - Suspicious activities were not detected early.
- No risk assessment before festive sale
 - High traffic period requires stronger security, which was ignored.

(ii) Role of vulnerability identification

Attackers scanned the website to find weaknesses such as:

- weak login systems
- unpatched software
- insecure APIs or plugins

These vulnerabilities acted as entry points. Proper vulnerability assessment could have identified and fixed them earlier.

Role of timing strategy

Attackers targeted the website during a festive sale because:

- Traffic was very high
- Security team was busy handling customers
- System overload reduced monitoring efficiency

High business dependency increased chances of financial loss and price

Conclusion

Poor security planning, failure to identify vulnerabilities, and attacker timing strategy together made the attack successful. Regular updates, strong password policy, and pre-event security testing could have prevented the breach.

Q2

A company experienced a DDOS attack during a public product launch event. Analyze why attackers might choose specific timing and attack methods. Compare the impact of poor preparation versus strong cybersecurity planning.

A DDOS attack is designed to flood a system with traffic so that legitimate users cannot access services.

(i) Why attackers choose specific timing

- Attackers strategically launch DDOS attacks during important events because:
- Product launch bring maximum traffic
 - Service downtime causes financial loss and reputational damage
 - Media attention increases the impact
 - Companies are under pressure and more likely to react emotionally.

(ii) Why specific attack methods are chosen

- DDOS attacks are effective because:

- They require no direct system access
- Large bot networks can overwhelm servers easily
- Difficult to distinguish fake traffic from real users.

Impact of poor preparation

- Website crashes or become unavailable
- Loss of customer trust
- Revenue loss during launch
- Slow recovery due to lack of incident response plan
- Negative publicity

iv) Impact of strong cybersecurity planning

- Use of DDoS protection services and firewalls
- Traffic filtering and load balancing
- Backup servers and cloud scaling
- Real-time monitoring and quick response
- Business continuity with minimal downtime

Conclusion

Attackers exploit critical business moments to maximize damage. Proper cybersecurity planning significantly reduces the impact and ensures service availability.
